

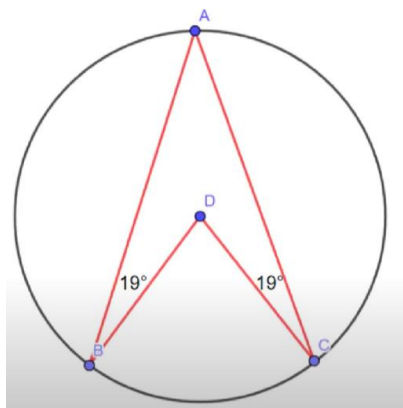
Hard questions 4

1.

$$x^2 - 12x + 7 = 0$$

One solution to the given equation can be written as  $j - \sqrt{k}$ , where  $j$  and  $k$  are constants. What is the value of  $jk$ ?

2.



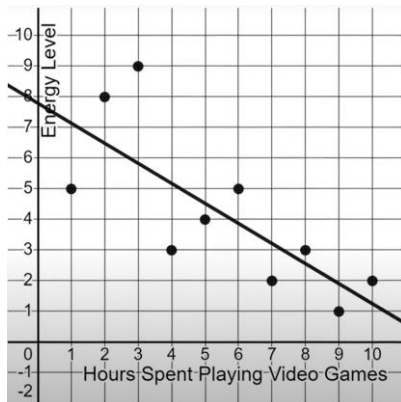
The measure of angle BDC, in radians, is  $\frac{a\pi}{b}$  where  $a$  and  $b$  are positive integers. What is the smallest possible value of  $a + b$ ?

3.

$$f(x) = |8 - 5x|$$

The function  $f$  is defined by the given function. For what value of  $k$  is  $f(k)$  equal to  $6k + 3$ ?

4.



The line of best fit underestimates one student's energy level by more than 3 points. What was this student's reported energy level?

5.

$$K = \frac{RT^2M}{1000H}$$

Solve for T.

A)  $T = \frac{1000^2 H^2 K^2}{R^2 M^2}$

B)  $T = 10\sqrt{\frac{10HK}{RM}}$

C)  $T = \sqrt{1000HK - R - M}$

D)  $T = 1000\sqrt{\frac{HK}{RM}}$

6.

$\sin(a^\circ) = \cos(b^\circ)$  where  $a$  and  $b$  are positive constants. The number  $a$  is 70% greater than the number  $b$ . What is the value of  $a$ ?

7.

For groups of 28 or more people, an arcade charges \$9 per person for the first 28 people and \$15 for each additional person. How many total people were in attendance if the group paid \$447 to go to the arcade?